

Mathematics Section A

1. For the function $f: [1, \infty) \rightarrow [1, \infty)$ defined by $f(x) = (x-1)^4 + 1$, among the two statements: (I) The set $S = \{x \in [1, \infty) : f(x) = f^{-1}(x)\}$ contains exactly two elements, and (II) The set $S = \{x \in [1, \infty) : f(x) = f^{-1}(x+1)\}$ is an empty set,

(A) only (I) is TRUE (B) only (II) is TRUE
(C) both (I) and (II) are TRUE (D) neither (I) nor (II) is TRUE
2. Let $S = \{z \in \mathbb{C} : z^2 + 4z + 16 = 0\}$. Then $\sum_{z \in S} |z + \sqrt{3}i|^2$ is equal to:

(A) 42 (B) 23 (C) 27 (D) 38
3. If the system of equations:

$$\begin{aligned} x + y + z &= 5 \\ x + 2y + 3z &= 9 \\ x + 3y + \lambda z &= \mu \end{aligned}$$
 has infinitely many solutions, then the value of $\lambda + \mu$ is:

(A) 16 (B) 18 (C) 19 (D) 21
4. If $\alpha = 1$ and $\beta = 1 + i\sqrt{2}$, where $i = \sqrt{-1}$ are two roots of the equation $x^3 + ax^2 + bx + c = 0$, $a, b, c \in \mathbb{R}$, then $\int^1 (x^3 + ax^2 + bx + c) dx$ is equal to:

(A) -2 (B) -4 (C) -8 (D) -10
5. If the quadratic equation $(\lambda + 2)x^2 - 3\lambda x + 4\lambda = 0$, $\lambda \neq -2$, has two positive roots, then the number of possible integral values of λ is:

(A) 1 (B) 2 (C) 3 (D) 4
6. Let $A = \begin{bmatrix} 1 & 2 & 7 \\ 4 & -2 & 8 \\ 3 & 8 & -7 \end{bmatrix}$ and $\det(A - \alpha I) = 0$, where α is a real number. If the largest possible value of α is p , then the circle $(x-p)^2 + (y-2p)^2 = 320$, intersects the co-ordinate axes at

(A) 1 point (B) 2 points (C) 3 points (D) 4 points
7. Let $\alpha = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \infty$ and $\beta = \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots \infty$. Then the value of $(0.2)^{\log_{\sqrt{5}}(\alpha)} + (0.04)^{\log_5(\beta)}$ is equal to:

(A) 4 (B) 5 (C) 8 (D) 25
8. For 10 observations $x_1, x_2, \dots, x_{10}$, if $\sum_{i=1}^{10} (x_i + 2)^2 = 180$ and $\sum_{i=1}^{10} (x_i - 1)^2 = 90$, then their standard deviation is:

(A) 2 (B) $\sqrt{3}$ (C) $2\sqrt{2}$ (D) 3
9. In the expansion of $(9x - \frac{1}{3\sqrt{x}})^{18}$, $x > 0$, if the term independent of x is $(221)k$, then k is equal to:

(A) 84 (B) 78 (C) 168 (D) 198
10. Let $P(3 \cos \alpha, 2 \sin \alpha)$, $\alpha \neq 0$, be a point on the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$, Q be a point on the circle $x^2 + y^2 - 14x - 14y + 82 = 0$ and R be a point on the line $x + y = 5$ such that the centroid of the triangle PQR is $(2 + \cos \alpha, 3 + \frac{2}{3} \sin \alpha)$. Then the sum of the ordinates of all possible points R is:

(A) 6 (B) 2 (C) 4 (D) 8
11. Let $H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be a hyperbola such that the distance between its foci is 6 and the distance between its directrices is $\frac{8}{3}$. If the line $x = \alpha$ intersects the hyperbola H at the points A and B such that the area of the triangle AOB is $4\sqrt{15}$, where O is the origin, then α^2 equals

(A) 12 (B) 16 (C) 24 (D) 25
12. $\max_{0 \leq x \leq \pi} (16 \sin(\frac{x}{2}) \cos^3(\frac{x}{2}))$ is equal to:

(A) $\frac{3\sqrt{3}}{2}$ (B) $3\sqrt{3}$
(C) $4\sqrt{3}$ (D) $6\sqrt{3}$
13. $\vec{r} = (\frac{1}{3}\hat{i} + 2\hat{j} + \frac{8}{3}\hat{k}) + \lambda(2\hat{i} - 5\hat{j} + 6\hat{k})$
and $\vec{r} = (-\frac{2}{3}\hat{i} - \frac{1}{3}\hat{k}) + \mu(\hat{j} - \hat{k})$, $\lambda, \mu \in \mathbb{R}$, is:

(A) $\sqrt{5}$ (B) 3 (C) $2\sqrt{3}$ (D) $\sqrt{15}$
14. If $(2\alpha + 1, \alpha^2 - 3\alpha, \frac{\alpha-1}{2})$ is the image of $(\alpha, 2\alpha, 1)$ in the line $\frac{x-2}{3} = \frac{y-1}{2} = \frac{z}{1}$, then the possible value(s) of α is (are)

(A) Only 3 (B) Only 3 and -1
(C) Only 3, $\frac{1}{4}$ and -1 (D) Only 3 and $\frac{1}{4}$
15. Let $\hat{u}$ and $\hat{v}$ be unit vectors inclined at an acute angle such that $|\hat{u} \times \hat{v}| = \frac{\sqrt{3}}{2}$. If $\vec{A} = \lambda\hat{u} + \hat{v} + (\hat{u} \times \hat{v})$, then λ is equal to:

(A) $\frac{4}{3}(\vec{A} \cdot \hat{u}) - \frac{2}{3}(\vec{A} \cdot \hat{v})$
(B) $\frac{2}{3}(\vec{A} \cdot \hat{u}) - \frac{1}{3}(\vec{A} \cdot \hat{v})$
(C) $\frac{4}{3}(\vec{A} \cdot \hat{u}) + \frac{2}{3}(\vec{A} \cdot \hat{v})$
(D) $(\vec{A} \cdot \hat{u}) - \frac{1}{2}(\vec{A} \cdot \hat{v})$
16. Let for some $\alpha \in \mathbb{R}$, $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying $f(x+y) = f(x) + 2y^2 + y + \alpha xy$ for all $x, y \in \mathbb{R}$. If $f(0) = -1$ and $f(1) = 2$, then the value of $\sum^5 (\alpha + f(n))$ is:

(A) 110 (B) 140 (C) 150 (D) 170
17. Let

$A = \{(a, b, c) : a, b, c \text{ are non-negative integers and } a + b + 2c = 22\}$. Then $n(A)$ is equal to:

- (A) 121 (B) 124 (C) 144 (D) 169

18. The area of the region bounded by the curves $x + 3y^2 = 0$ and $x + 4y^2 = 1$ is equal to:

- (A) $\frac{1}{3}$ (B) $\frac{2}{3}$
(C) $\frac{4}{3}$ (D) $\frac{5}{3}$

19. Let $y = y(x)$ be the solution of the differential equation:

$$\frac{dy}{dx} + \left(\frac{6x^2 + (3x^2 + 2x^3 + 4)e^{-2x}}{(x^3 + 2)(2 + e^{-2x})} \right) y = 2 + e^{-2x},$$

$x \in (-1, 2)$, satisfying $y(0) = \frac{3}{2}$. If $y(1) = \alpha(2 + e^{-2})$, then α is equal to:

- (A) $\frac{13}{8}$ (B) $\frac{6}{13}$
(C) $\frac{12}{13}$ (D) $\frac{13}{12}$

20. The integral $\int_0^1 \cot^{-1}(1 + x + x^2) dx$ is equal to:

- (A) $2 \tan^{-1} 2 + \frac{1}{2} \log_e \left(\frac{5}{4} \right) + \frac{\pi}{2}$
(B) $2 \tan^{-1} 2 + \frac{1}{2} \log_e \left(\frac{5}{4} \right) - \frac{\pi}{2}$
(C) $2 \tan^{-1} 2 - \frac{1}{2} \log_e \left(\frac{5}{4} \right) + \frac{\pi}{2}$
(D) $2 \tan^{-1} 2 - \frac{1}{2} \log_e \left(\frac{5}{4} \right) - \frac{\pi}{2}$

Mathematics Section B

21. From a month of 31 days, 3 different dates are selected at random. If the probability that these dates are in an increasing A.P. is equal to $\frac{a}{b}$, where $a, b \in \mathbb{N}$ and $\gcd(a, b) = 1$, then $a + b$ is equal to ____

22. Let $f(x) = \begin{cases} e^{x-1} & , x < 0 \\ x^2 - 5x + 6 & , x \geq 0 \end{cases}$ and $g(x) = f(|x|) + |f(x)|$. If the number of points where g is not continuous and is not differentiable are α and β respectively, then $\alpha + \beta$ is equal to ____

23. Let A, B be points on the two half-lines $x - \sqrt{3}|y| = \alpha, \alpha > 0$ at a distance of α from their point of intersection P. The line segment AB meets the angle bisector of the given half-lines at the point Q. If $PQ = \frac{9}{2}$ and R is the radius of the

circumcircle of $\triangle PAB$, then $\frac{a^2}{R}$ is equal to ____

24. Let A, B and C be the vertices of a variable right angle triangle inscribed in the parabola $y^2 = 16x$. Let the vertex B containing the right angle be $(4, 8)$ and the locus of the centroid of $\triangle ABC$ be a conic C_0 . Then three times the length of latus rectum of C_0 is ____

25. Let f be a twice differentiable function such that $f(x) = \int_x^x \tan(t-x) dt - \int_x^x f(t) \tan t dt, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then $f''\left(\frac{\pi}{6}\right) + 12f'\left(-\frac{\pi}{6}\right) + f\left(\frac{\pi}{6}\right)$ is equal to ____

Physics Section A

26.	List-I		List-II
A	Planck's constant	I	$\text{ML}^2 \text{T}^{-2}$
B	Stopping potential	II	T^{-1}
C	Work function	III	$\text{ML}^2 \text{T}^{-1}$
D	Threshold frequency	IV	$\text{ML}^2 \text{T}^{-3} \text{A}^{-1}$

- (A) A-III, B-IV, C-I, D-II (B) A-I, B-II, C-III, D-IV
(C) A-IV, B-III, C-I, D-II (D) A-I, B-IV, C-III, D-II

27. Two cars A and B are moving in the same direction along a straight line with speeds 100 km/h and 80 km/h, respectively such that car A is moving ahead of car B. A person in car B throws a stone with a speed v so that it hits the car A with a speed of 5 m/s. The value of v is ____ km/h.

- (A) 18 (B) 28 (C) 38 (D) 48

28. At $t = 0$, a body of mass 100 g starts moving under the influence of a force $(5\hat{i} + 10\hat{j})\text{N}$. After 2 s its position is $(2x\hat{i} + 5y\hat{j})\text{m}$. The ratio $x : y$ is ____

- (A) 1 : 2 (B) 2 : 5 (C) 5 : 2 (D) 5 : 4

29. If x and y coordinates of a projectile as a function of time (t) are given as $24t$ and $43.6t - 4.9t^2$, respectively, the angle (in degrees) made by the projectile with horizontal when $t = 2$ s is ____.

- (A) 60 (B) 45 (C) 30 (D) 75